

A Quantitative Comparison of the Hominoid Thalamus

I. SPECIFIC SENSORY RELAY NUCLEI

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ABSTRACT Studies to date have indicated few differences in sensory perception among hominoids. Sensory relay nuclei in the dorsal thalamus—portions of the medial and lateral geniculate bodies (MGBp, LGBd) and the ventrobasal complex (VB)—in two gibbons, one gorilla, one chimpanzee and three humans were examined for anatomical similarity by measuring and estimating the nuclear volumes, neuronal densities, numbers of neurons per nucleus, and volumes of neuronal perikarya. The absolute volumes of these nuclei were larger in the larger brains; however, with the volume of the dorsal thalamus as a standard, these sensory relay nuclei showed negative allometry. The gibbons had about half as many neurons as did the other hominoids. Although the human VB had slightly more neurons, the numbers of neurons in LGBd and MGBp did not significantly differ between the great apes and humans. The volumetric distribution of the neuronal perikarya were similar among these hominoids. Other thalamic nuclei had much more diverse numbers of neurons and relative frequencies of their neuronal perikarya. The sensory relay nuclei appear to be a group of conservative nuclei in the forebrain. These results suggest that as a neurological base for complex behaviors evolved in hominids, not all parts of the brain changed equally.

Apes and humans appear to differ only slightly in primary sensory perception (Elder, '34; Farrer and Prim, '65; Farrer and Young, '70; Grether, '41, '42; Kellogg and Kellogg, '33; Kohler, '15; Riesen, '70; Spence, '34). Consequently, among extant hominoids, areas of the brain directly associated with these functions should be histologically similar. In the thalamus, several sensory areas can be compared within a limited region of the brain, thus providing information of both absolute and relative differences. In this pilot study to test for structural similarity of sensory nuclei among hominoids, measurements from the medial and lateral geniculate bodies and the ventrobasal complex in two gibbons, one gorilla, one chimpanzee, and three humans were compared.

The following brief summary of information on the sensory nuclei of the dorsal thalamus is primarily based on findings in primates. When

primate data are lacking, data from other specified mammals are given.

Medial geniculate body

The medial geniculate body (MGB), in the ventral and posterior part of the thalamus (fig. 1), is subdivided into a small pars magnocellularis (MGBm) and a large pars parvocellularis (MGBp). Only the MGBp, the specific auditory relay nucleus of the thalamus, was examined in this study.

All MGBp neurons respond to tones, but the precise role played by MGBp is not yet known. A tonotopic organization of synapses has been detected in MGBp in the squirrel monkey (Gross et al., '74). In cats, MGBp may complete frequency analyses that are partially accomplished in lower nuclei of the auditory sys-

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tem (Katsuki, '66). It may also be involved with discriminating sound patterns and localizing sound in space (Neff et al., '72). Neurons responsive to patterns of sinusoidal waves, to changes in stimulus frequencies, and to rising and falling frequencies are located here (Keidel, '74). Keidel hypothesized that those three electrophysiological patterns correspond to vowels, consonants, and transient detectors respectively.

Lateral geniculate body

The lateral geniculate body (LGB) is composed of two components, LGB ventralis and LGB dorsalis. Because only the latter component was studied, LGB in this article refers to LGB dorsalis.

The LGB is surrounded by fibers and has a distinct laminar arrangement internally (figs. 1, 2). The basal two layers (numbered 1 and 2), the laminae around the hilum, are known as the pars magnocellularis (LGBm). The other layers, composed of only small cells are called pars parvocellularis (LGBp) (fig. 1). The number of laminae within LGBp varies among primates (Giolli and Tigges, '70; Noback and Laemle, '70). While the synapses in the parvo- and magnocellular regions are basically similar (Glees, '61), LGBm may have slightly large axonal arborizations (Tigges and O'Steen, '74).

The LGB is the specific visual relay nucleus of the thalamus. First-order projections from the retinal ganglion cells synapse in LGB. The visual projections are visuotopic (Le Gros Clark, '32).

The LGB modifies some of the retinal input, but its precise functional roles are not known. Each LGB neuron appears to enhance contours (De Valois and Pease, '71), respond to the movement of light in all directions (Campbell, '74), and respond either to all wavelengths (white light) or differentially to restricted wavelengths (color) (Weisel and Hubel, '66). Activity of LGB may frame or punctuate visual input and increase the capacity to transmit patterns to the cortex (McIlwain, '72; Doty et al., '73). Synapses at the LGB level may provide for a constancy of depth perception during disjunctive eye movements (Richards, '68).

Ventrobasal complex

The ventrobasal complex (VB) has two nuclei, the ventroposterior lateralis (VPL) and medialis (VPM). The rostral region of VB is

sometimes segregated as the nucleus ventrointermedius (VI) because it is composed solely of large cells, (Hassler, '59; Van Buren and Borke, '72; Kanagasuntheram and Wong, '68; Andrew and Watkins, '69), but in this study it is included within VB because occasional areas of large cells without medium and small cells were found not only rostrally, but also in pockets throughout lateral VB. However, a small-celled nuclear region medial to VPM is distinguished from the magnocellular portion of VPM by its lack of large neurons. Although the same bundle of myelinated fibers surrounds both the large- and small-celled regions, only the portion containing large neurons was analyzed in this study. A similar division was made by others (Dekaban, '53; Hassler, '59; Shantha and Manocha, '69; Van Buren and Borke, '72). The lightly stained, small-celled region, nucleus ventroposterior inferior (VPI), was also not examined in this study (fig. 3).

The VB relays information to the cerebral cortex, primarily from mechanoreceptors (touch, pressure) as well from thermal, kinesthetic, and possibly pain receptors.

Classically, VB is seen as a link in the mammalian medial lemniscal system and thus is concerned with spatially localized, rapidly adapting light touch that is involved in tasks requiring discriminative touching (Welker, '73). Stimulation of VB in humans produces subjective sensations of tingling, "pins and needles," or numbness in the contralateral face and body (Mark et al., '63; Albe-Fessard, '73). These sensations are highly localized, but nonspecific in modality (Mark and Ervin, '60). Surgical lesions in VB give mediocre, if any, relief from intractable pain and lead to unpleasant sensations, ill-defined in time and space (Fazio et al., '73; Mark et al., '63). The VB may function to gauge or control the interaction of information arriving along several convergent somatosensory pathways rather than be involved directly in conscious sensory mechanisms (Wall and Dubner, '72).

MATERIALS AND METHODS

Specimens

The thalami of hominoids were studied quantitatively in specimens from one gorilla, one chimpanzee, two gibbons, and three humans.

The gorilla's thalamus, obtained from the Department of Anatomy at Columbia University, was from a 14-year-old male mountain

gorilla (*Gorilla gorilla*) weighing 203 kg, with a brain weight of 540 gm. The body of the animal, which had drowned in the moat at the Bronx Zoo, was refrigerated overnight and the brain removed within 24 hours after the time of death. The brain was then fixed by immersion in 10% formalin, embedded in celloidin, and cut into coronal serial sections, 30 μ m thick. Every tenth section was stained with cresyl violet (Gurr). In another series of sections, myelin sheaths were stained by hematoxylin using the Mahon methods.

The chimpanzee brain, from the comparative brain collection in the Laboratory of Neurophysiology at the University of Wisconsin, was from a 63.5-kg male chimpanzee (*Pan troglodytes*), aged 10 to 12 years. The 405.5-gm brain was perfused with saline and then with formalin, hardened in celloidin, and sectioned coronally at 40 μ m. Every fifth section was stained with thionin. A myelin-stained series was also available.

Thalami from the two hemispheres of one brain, from a gibbon (*Hylobates*) of unknown species and weight, were from the department of anatomy at Thomas Jefferson Medical College. The two thalami were fixed in 10% formalin, embedded in celloidin, and serially sectioned at 25 μ m in the horizontal and sagittal planes. Every fifth section was stained with cresyl violet. A myelin-stained series was also available.

A third gibbon brain specimen was from the Max-Planck-Institute of Frankfurt, which had imported the presumably wild male gibbon (*Hylobates lar*) from southeast Asia. The animal's body weight was 5,930 gm. The Bouin's-fixed brain, weighing 101.9 gm, was embedded in paraffin and serially sectioned in the transverse (coronal) plane. Every twelfth section was 20 μ m thick and stained with cresyl violet. A myelin-stained series (Heidenhain-Woelcke) was also available.

Two normal human brains came from the Yakovlev Collection at the Armed Forces Institute of Pathology in Washington, D.C. One thalamus was from a man, aged 28 years, who died shortly after fracturing his neck in a fall. The second was from a 32-year-old woman who died of a thoracic aneurysm. Pathologists had found both brains normal. The brain weights were 1,890 gm (after fixation in 10% formalin) and 1,200 gm (fresh) respectively. Both brains were fixed in 10% formalin, embedded in celloidin, and sectioned 35 μ m thick. The male brain was sectioned sagittally, and every

tenth section was stained with cresyl violet. A myelin stain (Loyez modification of Weigert) was also provided. These thalami were used to identify the thalamic nuclei of interest and to measure nuclear volumes.

A third human brain, used for measuring cell sizes and counting cells, was from a 113-kg, 19-year-old man who died of aspiration of vomitus. The brain, weighing 1,500 gm (fresh), was fixed in 10% formalin, embedded in celloidin, sectioned 30 μ m thick, and stained with cresyl violet. The brain could not be used for the volumetric part of the study because part of the LGB had been cut.

The mean brain weight (1,545 gm) of the first two human brains approximated that of the third human brain (1,500 gm). Calculations of the number of neurons per nucleus combine nuclear volume and neuronal density. If neuronal density varied according to the size of the nucleus, an adequate estimation of the number of neurons should be found by taking the mean nuclear volume from the first two brains and combining this volume with the neuronal density from the third brain. The consistency of this methodology was tested in the anterior nucleus. The differences in estimated numbers of neurons was less than 5% when the number of neurons estimated for the third brain was compared with the number of neurons estimated from the combined value. So slight a difference can be attributable to individual variation.

For quick reference, each type of specimen was given a simplified code name, as listed in parenthesis: gorilla (*Gorilla g.*); chimpanzee (*Pan t.*); gibbon specimen cut on a horizontal plane (*Hylo.-h.*); gibbon specimen cut on a sagittal plan (*Hylo.-s.*); the third gibbon specimen, cut transversely (*H. lar-t.*); the human male brain, cut sagittally (*Homo s.-s.*); the human female brain, cut transversely (*Homo s.-t.*); and the third human brain, used for cell count (*Homo s.-c.*).

Nuclear volume

The quantitative techniques were similar for all specimens. The boundaries of each nucleus were determined at a relatively low magnification of 15-21.6 $\times$. The image of each section was projected by means of a flat-field optical system (with planar optics) onto a piece of paper, and the outlines of the nuclei were traced. Thalamic nuclei, unlike the cortex, can be observed and measured in all three spatial dimensions. The area of each outlined

nucleus was determined by following the outline with the tracer lens of a compensating polar planimeter. Two planimetric tracings of each outline were made at two arm positions. This precautionary check on the reliability of the measurements was made because of the irregularity of the nuclear shapes. The volume (V) of the nucleus is the sum of the area (A), times the section thickness (t), multiplied by the distance between tracings (d), and divided by the magnification of each axis (M^2):

$$V = \frac{A(t)(d)}{M^2}.$$

The measurements of the nuclei were not sufficiently precise to demand a more elaborate formula for estimating the volume (Bauchot, '63). The volume of a nucleus from each specimen was calculated from the measurement of only one hemisphere. The measurements for LGB included the fibrous laminae between the cellular regions.

The thalami in this study came from different laboratories. Different preparative techniques can produce variable amounts of tissue shrinkage (Frontera, '58; Stephan et al., '70). An assumption of this study is that shrinkage within the thalamus is approximately the same for all regions (Stephan et al., '70). Comparisons that rely on ratios of the nucleus to the larger thalamus either directly or through an allometric analysis are comparisons that control for shrinkage.

Cell size

Neuronal density and the sizes of neuronal perikarya were estimated. Every twenty-fifth section from *Homo s-c* and *Pan t.* was sampled, every twentieth from *Gorilla g.*, every thirty-sixth from *H. lar.* and every fifteenth from *Hylo-S* and *-h*. Crossbars on a grid sketched on the tracings of the entire thalamic nucleus being measured marked the sampled areas of the nuclei.

Only the neuronal perikarya that had observable nuclei and nucleoli were measured. Based on the premise that the neuron has an approximately elliptical shape, a frequent, but not universal condition (Van Harreveld and Schade, '62), the perikaryal volume was calculated by the formula:

$$V = 1.04 \times 1/6 \pi (ab \sqrt{ab}).$$

V is the volume of the soma, a is the greatest length of the perikaryon passing through the nucleolus, and b is the greatest width of the

perikaryon passing through the nucleolus at right angles to a . These axes were measured with an eye-piece micrometer under high magnification ($860 \times$). The data thus accumulated are considered to be tentative because of several factors: the sample size was small; not all parts of the nucleus were sampled; and the histological processing was different for each specimen. The information is presented as an addition to similar studies (Kraus and Gihl, '65; Blinkov and Glezer, '68) to help establish the range of variation to be expected.

Cell density

Neuronal density was sampled in fields delimited by a subdivided square reticule mounted in the eyepiece. Counting procedures followed those of Dornfield et al. ('42), Konigsmark ('70), and Welker et al. ('64). Only those neurons with visible nucleoli were counted. When the outside edges of the reticule exactly bisected the nucleolus of a cell, that cell was counted only if it was above or to the right edge of the reticule. That procedure was a means of avoiding counting the cells twice when neurons in adjacent portions of a cellular region were counted. Two to four counts were made over the same microscope field.

Estimates of the total number of neurons per nucleus were then obtained by the following formula:

$$N_t = \frac{N_s}{V_s} V_t.$$

N_t is the estimated total number of neurons, N_s is the sampled number of neurons, V_s is the sampled volume, and V_t is the total volume of the nucleus.

The estimates of neuronal density were not corrected according to the size of the neuronal nucleus.² These brains were embedded in relatively soft material (paraffin and celloidin). Nucleoli are not frequently split in this condition (Cammermeyer, '67; Jones, '37; Konigsmark, '70).

Neuronal density is also affected by tissue shrinkage. Greater shrinkage decreases the volume of tissue and increases neuronal density. If the density figures are combined with the volume of tissue the result gives the number of neurons per nucleus. The number of neurons per nucleus is a ratio that can com-

²The numbers of neurons calculated with the Abercrombie correction factor are not necessarily more correct. I have cited some of the references involved in this dispute. Calculations made in the gorilla and human anterior nucleus showed that the correction factor did not change the relative difference between these specimens.

TABLE 1

Volumes of specific sensory relay nuclei: LGB, MGBp, VB

Specimen	LGB volume in mm ³	% of total thalamus	MGBp volume in mm ³	% of total thalamus	VB volume in mm ³	% of total thalamus
<i>This study</i>						
Hylo.-s	17.8	4.8	7.1	1.9	41.2	11.2
Hylo.-h	21.0	5.6	8.3	2.2	37.5	10.2
H. lar-t	27.1	4.7	11.8	2.1	62.4	10.8
Gorilla g.	51.0	3.2	29.0	1.8	178.9	11.2
Pan t.	56.3	3.8	28.0	1.9	144.8	9.9
Homo s.-s	69.1	1.6	71.2	1.7	300.3	7.1
Homo s.-t	78.0	2.3	46.6	1.4	273.1	8.1
<i>Hopf ('65)</i>						
Pongo p.	165	4.0	50	1.1		
Homo S.	160	1.7	105	1.1		
<i>Solnitzky ('45)</i>						
Pan t.	61.9					
Homo s.	125.2					
<i>Chacko ('48)</i>						
Homo s.	60.6					
<i>Blinkov and Zvorykin ('50)</i>			MGP p + m			
Hyllobates			12.2			
Pongo			33.9			
Pan			36.5			
Homo s. (mean of 10 hemispheres)			67.9			

pensate for variable shrinkage (Powell et al., '57) and is thus the more important datum since it is not biased by density variations.

Cytoarchitecture

The neuronal architecture was basically similar among the hominoids examined. Neurons in MGBp were round to multipolar and took a medium stain. The neurons in LGBp and LGBm were more darkly stained. The large neurons in LGBm stained most heavily. In all the nuclei studied, the neuronal somata were predominantly round. Size and composition of VB neurons, however, varied—some were round, some multipolar. Whereas the larger neurons in the ventrobasal complex were consistently dark, staining of the medium and small neurons varied from light to dark.

The laminar appearance of LGB differed among the hominoids. The basal two layers of LGBm were similar, but LGBp of humans and chimpanzees had four laminae whereas the gibbons had just two (figs. 1, 2).

Nuclear volume

Table 1 shows the total volume of each of the sensory nuclei. For all the sensory nuclei, the gibbons had the smallest volumes and the humans, the largest. The great apes and hu-

mans differed least in the absolute sizes of their LGB. The gorilla specimen had larger absolute volumes in VB and MGBp than did the chimpanzee specimen.

The sensory nuclei occupied a larger proportion of the thalamus in the gibbons than in any of the other hominoids. The gorilla VB occupied more of the thalamus than did the VB in the chimpanzee, but the proportions of MGBp and LGB did not appear to differ significantly between the two.

Although the volumes of these relay nuclei are greater in the larger thalami, they constitute a smaller proportion of the larger thalami. The proportional decrease differentiates the relay nuclei from other thalamic nuclei. The absolute differences in volumes between species are substantive, but are not as great as those among the other thalamic nuclei. Disproportionate increase and decreases can, of course, be expected when animals of different sizes are compared. Such differential variations are particularly relevant when the ratios of a surface structure, such as the cerebral cortex, are compared with those with a cubic structure, such as the total brain. In the case of the dorsal thalamus all but one ³ of the nuclei are cubic and change size as such.

³The exception is the N reticularis, which encapsulates the thalamus on the dorsal and lateral surfaces.

TABLE 2

Neuronal density: MGB parvocellularis

Specimen	No. of fields	Mean No. of neurons per 0.01 mm ²	S.D.	S.E.	Minimum	Maximum	Median
Hylo.-s	13	532.1	108.9	30.2	354.2	777.8	503.5
Hylo.-h	14	479.0	142.5	38.1	222.2	676.2	518.0
H. lar-t	16	345.1	89.0	22.1	193.3	608.3	330.8
Gorilla g.	24	243.9	82.9	16.9	64.7	418.7	245.1
Pan t.	14	325.9	91.3	24.4	224.4	495.6	287.0
Homo s.-c	30	147.7	42.6	7.8	80.0	305.0	142.0

TABLE 3

Neuronal density: LGB

Specimen	No. of fields	Mean No. of neurons per 0.01 mm ²	S.D.	S.E.	Minimum	Maximum	Median
LGBp							
Hylo.-s	20	784.7	172.5	38.6	458.3	1,083.3	781.3
Hylo.-h	30	755.5	174.2	31.8	291.7	1,000.0	809.0
Gorilla g.	28	385.3	82.6	15.6	217.6	629.4	370.6
Pan t.	21	397.7	73.6	16.1	264.0	540.0	405.5
Homo s.-c	31	303.9	62.3	11.1	175.0	415.0	313.8
LGBm							
Hylo.-s	7	368.0	119.0	45.0	243.1	579.2	366.3
Hylo.-h	19	382.7	119.9	27.5	250.0	652.8	343.8
Gorilla g.	10	288.8	121.5	38.4	170.6	506.9	253.4
Pan t.	11	350.2	80.7	24.3	236.0	516.0	346.0
Homo s.-c	14	181.8	51.4	13.7	120.0	295.0	170.0

TABLE 4

Neuronal density: VB

Specimens	No. of fields	Mean No. of neurons per 0.01 mm ²	S.D.	S.E.	Minimum	Maximum	Median
Hylo.-s	43	142.5	57.0	8.7	55.6	263.9	138.9
Hylo.-h	30	122.1	34.8	6.4	69.4	222.2	116.3
H. lar-t	40	123.4	32.9	5.2	61.7	213.3	124.6
Gorilla g.	61	60.0	26.0	3.3	24.4	129.4	53.8
Pan t.	33	84.7	29.1	5.1	36.0	144.0	76.3
Homo s.-c	48	58.8	22.0	3.4	25.0	115.0	57.7

In this phylogenetic series, a log.-log. plotting of the volumes of each nucleus against the volume of the dorsal thalamus minus the nuclear volume produces the lines of highest correlation (fig. 6). The relationship of the size of each nucleus to that of the total thalamic volume was determined by the allometric formula $\log Y = \alpha(\log X) + \log b$ (Gould, '66). Unlike most of the other thalamic nuclei, all the sensory nuclei have slopes that are less than one, an indication that these nuclei have experienced a differential decline in hominoids. This allometric interpretation is supported by the finding of a similar number of

neurons in the sensory nuclei of great apes and humans.

Neuronal density

The neuronal density was least in the human specimen and greatest in the gibbons (tables 2-4). The great apes had intermediate values. All of the gorilla nuclei, even in LGBm where the total volume of the nucleus was smaller than that of the chimpanzee, had less dense neuronal packing than did those of the chimpanzee (table 5).

The numbers of neurons in MGBp and LGB were similar among the great apes and hu-

mans (table 6). The VB in the human specimens had more neurons than did the VB in the great apes. The gibbons had about 50% fewer neurons in VB than did any of the other hominoids studied.

Neuronal volume

Few consistent differences in the distribution of perikaryal volumes were found. A slightly more differentiated neuronal population was observed in the human MGBp.

The estimates of the volumes of neuronal perikarya are listed in tables 7-9. Because these volumes vary widely, analyzing a single statistic, such as the mean, is less informative than comparing the relative frequencies (i.e., shapes of the distributions) of the perikaryal volume. The shapes of the distributions of specific sensory relay nuclei are similar among

the hominoids measured (figs. 4, 5), except for slightly smaller range of perikaryal volumes in the MGBp and LGB of the gibbons (fig. 4).

DISCUSSION

In the present study, the volumes of the three specific sensory relay nuclei, LGB, MGBp, and VB, were found to be largest in the humans and smallest in the gibbons, with the greatest differences in volumes between the gibbons and the rest of the hominoids. Other investigators reported similar results (table 1). Solnitzky ('45) estimated the LGB to be larger in a human than in a chimpanzee, but Hopf ('65) described the absolute size of LGB as similar in a human and orangutan. A study of LGB volumes suggests that the absolute size of this nucleus may be progressive (as termed by Stephan, '69) in the primate order, but the data from my study show the differences among hominoids to be conservative.

The MGB in Hopf's orangutan specimen was smaller than that in his human specimen (Hopf, '65). In the investigation by Blinkov and Zvorykin ('50), as in my study, the volumes of MGB differed in absolute size among *Hylobates*, *Pongo*, *Pan*, and *Homo*. Those investigators also found that, in humans, MGB volume ranged from 50.6-78.6 mm³, with a standard deviation of 8.3 mm³ in 10 hemispheres (Blinkov and Zvorykin, '50). Assuming that a similar amount of variation is present in the MGB in the apes, the volumetric differences observed in the present study among the great apes could be due to individual variations. The other differences are probably not.

No other volumetric work on VB has been done in hominoids. Further documentation is needed.

Individual variations in volume within a species are to be expected (Welker, '73; Carlson and Welker, '76; Welker and Johnson,

TABLE 5

Volume of LGB

Specimen	Volume in mm ³ LGBp	% of total LGB	Volume in mm ³ LGBm	% of total LGB
Hylo.-s	11.4	64.1	6.4	35.9
Hylo.-h	14.2	67.7	6.8	32.3
H. lar-t	21.4	79.0	5.7	21.0
Gorilla g.	42.6	83.4	8.4	16.6
Pan t.	42.3	75.1	14.0	24.9
Homo s.-s	57.7	83.3	11.5	16.6
Homo s.-t	62.4	80.0	15.6	20.0

TABLE 6

Estimated number of neurons

Specimen	Estimated number of neurons		
	LGB	MGBp	VB
Hylo.-s	1,131,000	378,000	587,100
Hylo.-h	1,333,000	413,000	458,000
H. lar-t	—	407,000	770,000
Gorilla g.	1,884,000	707,000	1,070,000
Pan t.	2,172,000	913,000	1,226,300
Homo s.	2,065,000	870,000	1,690,000

TABLE 7

Neuronal volume: MGB parvocellularis

Specimen	N	Mean neuron vol. in μm^3	S.D.	S.E.	Minimum	Maximum	Median
Hylo.-s	65	1,203.6	757.0	93.9	179.1	4,747.3	1,050.4
Hylo.-h	70	1,217.0	570.1	68.1	333.1	3,095.7	1,095.0
H. lar-t	18	2,022.5	1,390.2	327.7	49.7	3,807.3	1,983.0
Gorilla g.	124	2,145.3	1,760.0	158.1	150.1	12,679.2	1,749.3
Pan t.	47	2,063.2	1,117.4	163.0	207.4	5,752.3	1,880.5
Homo s.-c	150	2,770.7	2,087.2	170.4	97.5	10,990.1	2,569.3

TABLE 8

Neuronal volume: LGB

Specimen	N	Mean neuron vol. in μm^3	S.D.	S.E.	Minimum	Maximum	Median
LGBp							
Hylo.-s	102	1,296.3	698.8	69.2	97.5	4,747.3	1,168.7
Hylo.-h	150	1,405.9	625.4	51.1	238.4	3,503.4	1,368.9
Gorilla g.	140	1,792.2	845.3	71.4	275.8	7,654.2	1,682.2
Pan t.	19	1,649.0	620.6	142.4	650.7	2,776.9	1,603.5
Homo s.-c	155	2,261.7	1,244.0	99.9	117.8	7,679.9	2,068.6
LGBm							
Hylo.-s	35	4,225.7	1,655.7	279.9	424.6	7,994.9	4,036.2
Hylo.-h	95	2,634.3	1,312.6	134.7	238.4	6,815.2	2,607.9
Gorilla g.	50	4,275.1	2,912.7	411.9	506.7	18,484.5	3,993.4
Homo s.-c	72	6,256.2	3,276.9	386.2	179.1	13,848.3	6,338.3

TABLE 9

Neuronal volume: VB

Specimen	N	Mean neuron volume in μm^3	S.D.	S.E.	Minimum	Maximum	Median
Hylo.-s	131	4,141.8	3,264.4	285.2	323.2	22,939.8	3,510.2
Hylo.-h	155	3,696.2	2,956.1	237.4	179.1	13,848.3	2,898.1
H. lar-t	45	4,424.3	4,160.2	620.2	49.1	17,043.5	3,286.4
Gorilla g.	306	5,912.6	5,937.1	339.4	53.1	38,871.9	3,889.4
Homo s.-c	238	4,305.9	4,396.7	285.0	139.6	26,293.7	3,075.1

'65). For example, the area of VB representing the thumb is likely to be larger, both relatively and absolutely, in humans. If one part of MGBp, which has cytoarchitectonically distinct groups of neurons within it, is particularly important in the perception of speech, that region may be proportionately larger in humans than in other hominoids. Auditory tests in the chimpanzee revealed that they can hear slightly higher frequency sounds than can age-matched humans (Elder, '34; Farrer and Prim, '65). This factor and the perception of speech need further investigation.

The numbers of neurons in VB differed the most between the great apes and humans. Human VB was estimated to have one and one-half times as many neurons as do those of the great apes, a somewhat surprising finding because of the large body size of the great apes. Such a quantitative difference may reflect a different arrangement of connections at this level or a less dense arrangement of sensory receptors peripherally in the great apes. Tactile discrimination is the best studied of the functions subserved by the epicritic system relayed through VB (Semmes and Turner, '77; Welker, '73; Werner, '77). The Kelloggs' study (Kellogg and Kellogg, '33) did

not reveal any difference in tactile discrimination between humans and chimpanzees. Studies of nerve endings in the skin suggest a similarity in the form and distribution of organized nerve endings between humans and chimpanzees (Ford and Perkins, '70; Beigert, '71). Winklemann ('61) observed an apparent similarity in the number of nerves in the hair follicles between a gorilla and a human, but thought that the dermal nerve endings were fewer in the gorilla than in humans. This work was not quantitative. Montagna ('77) and Montagna and his associates (Montagna et al., '75) compared the dermal system of humans with those of other mammals and primates, but not other hominoids, and concluded that the "human skin is better equipped for cutaneous sensitivity than any other mammal we know (Montagna, '77: p. 4). The greater number of neurons in VB in the human than in the other hominoids of my study supports this contention.

An increase in the number of neurons in the human VB is not accompanied by changes in the relative frequencies of neurons when they are classified by their perikaryal sizes. In other thalamic nuclei, differences in the number of neurons are often accompanied by differences in the relative frequencies of neu-

ronal volumes (Armstrong, '76; and in preparation). This lack of alteration in neuronal proportions emphasizes the conservative nature of the changes in VB. Furthermore, the basic organization of the neurons may be similar among these hominoids as it is with other mammals. Although neither flattened nor tangential sections were available for study, coronal sections showed alignments of neurons suggestive of "barreloids" (Van Der Loos, '76) in these hominoids (fig. 3). The presence of "barreloids" in large hominoid thalami suggests that this type of organization of nerve cells is not confined to representing vibrissae at a thalamic level. The alignment may be important for general tactile organization or it may indicate locations for specialized tactile organization, e.g., in the case of hominoids, perhaps fingertips.

The estimated number of neurons per nucleus depends on the volume and neuronal densities. In general, larger brains have been found to have lower neuronal densities (Bok, '59; Shariff, '53; Tower, '54). This trend was noted among all the thalamic nuclei observed in this study (Armstrong, '76). However, the degree of difference in density varied with the nucleus being measured. An estimation of the number of neurons in the sensory relay nuclei revealed that in these, and only in these, thalamic nuclei, the great apes and humans have approximately the same number of neurons. The gibbons have about half as many neurons (table 6). The fewer neurons in the gibbons may well reflect their smaller body size. However, Beck ('67), in studying gibbon intelligence, noted that the gibbon seems to fixate on the wire mesh of the cage and not the test material. Although this figure-and-ground problem could be related to primary sensory perception, it more likely reflects the integration of visual perceptions. The quantitative differences in the number of neurons as well as the qualitative difference in LGB lamination pattern of gibbons should be more thoroughly explored, but until such studies are done, the fewer neurons in *Hylobates* are more appropriately interpreted as reflecting a smaller body size.

These data suggest that humans have fewer sensory neurons than would be expected in a nonhuman primate with the same thalamic volume. Although their brain sizes are equivalent (Tobias, '71), *Macaca mulatta* may have a volumetrically larger LGB than does the gibbon. Absolute LGB volume of a macaque was

29 cc (without laminae) (Le Gros Clark, '41), compared with 22 cc (with laminae) for the average LGB volume of the three gibbons in my study. This quantitative comparison needs verification because the specimens were prepared in different laboratories using different techniques. The macaque pattern of lamination appears more complex than that of any of the hominoids of this study, suggesting that the macaque LGB is more differentiated than not only that of the human, as Le Gros Clark ('41) concluded, but also those of all hominoids. That statement would be in agreement with the high placement of *Cercopithecidae* in Stephan's ('69) index of progression for LGB. Hominoids may have experienced less evolution in their visual nuclei than *Cercopithecidae*. The complexity of laminae in humans and macaques is probably the result of parallel evolution.

Selection pressures may have been working to stabilize the sensory nuclei during hominoid evolution. The fact that humans and great apes appear to have the same number of neurons in their visual and auditory relay nuclei supports behavioral studies that show a similarity in those systems. The similarity in MGB suggests that the neurological substrate for language lies elsewhere, probably in the expanded and differentiated association regions of the brain (Armstrong, '76; and in preparation). The auditory relay nuclei in the common ancestors of great apes and humans may have had sufficiently generalized functions to remain unchanged.

The increase in number of neurons in the human VB could be related to the importance of touching as opposed to grooming (Montague, '71) and/or to the increased development of tactile sensation necessary to support such cognitive skills as those needed in the making of tools. The fact that the human VB has more neurons than those of the other hominoids suggests some degree of differential evolution within the sensory regions of the brain. If such a difference is also found in the cortex, estimates of the numbers of "extra" cortical neurons (Jerison, '73) may need slight revisions.

Although the size of the human brain is impressive, size alone cannot account for human behavior (Holloway, '68). The largest increase in size in the cortex is thought to be in "association" regions (Holloway, '68; Passingham, '75). The brain is too complex to understand as a single unit. It does not have

one function, nor should its evolution be seen as responsive to one selection pressure. We should not be surprised that Old World monkeys may have a greater relative development in certain parts of their brains than do hominoids. The successful evolution of *Cercopithecidae* may have depended in part on the evolution of its visual system. The evolution of hominids may have depended on the stabilization of the retino-geniculate visual system and other sensory systems with the concomitant elaboration of integration and association regions. The evolution of integrative centers does not necessarily depend on the stabilization of sensory systems, but in human evolution that dependence appears to be the case.

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Abbreviations

Cd, Caudate nucleus	MGBp, Medial geniculate body, pars parvocellularis
CM, Nucleus centrum medianum	PC, Posterior commissure
FOR, Fornix	PF, Nucleus parafascicular
Hb, Habenular nucleus	Pr, Pulvinar
Hip, Hippocampus	Puo, Pulvinar oralis
ILA, Intralaminar nuclei	R, Nucleus reticularis
LD, Nucleus lateral dorsalis	Ru, Red nucleus
LGBd, Lateral geniculate body, pars dorsalis	SC, Superior colliculus
LGBm, Lateral geniculate body, pars magnocellularis	Sm, Stria medullaris
LGBp, Lateral geniculate body, pars parvocellularis	ST, Subthalamic nucleus
LGBv, Lateral geniculate body, pars ventralis	SN, Substantia nigra
Li, Nucleus limitans	teg, Tegmentum
LP, Nucleus lateral posterior	VB, Ventrobasal complex
LV, Lateral ventricle	VL, Ventrolateral nucleus
MD, Nucleus medial dorsalis	VPI, Nucleus ventroposterior inferior
MGBm, Medial geniculate body, pars magnocellularis	III, Third ventricle

PLATE 1

EXPLANATION OF FIGURE

- 1 Coronal section through posterior part of chimpanzee thalamus.

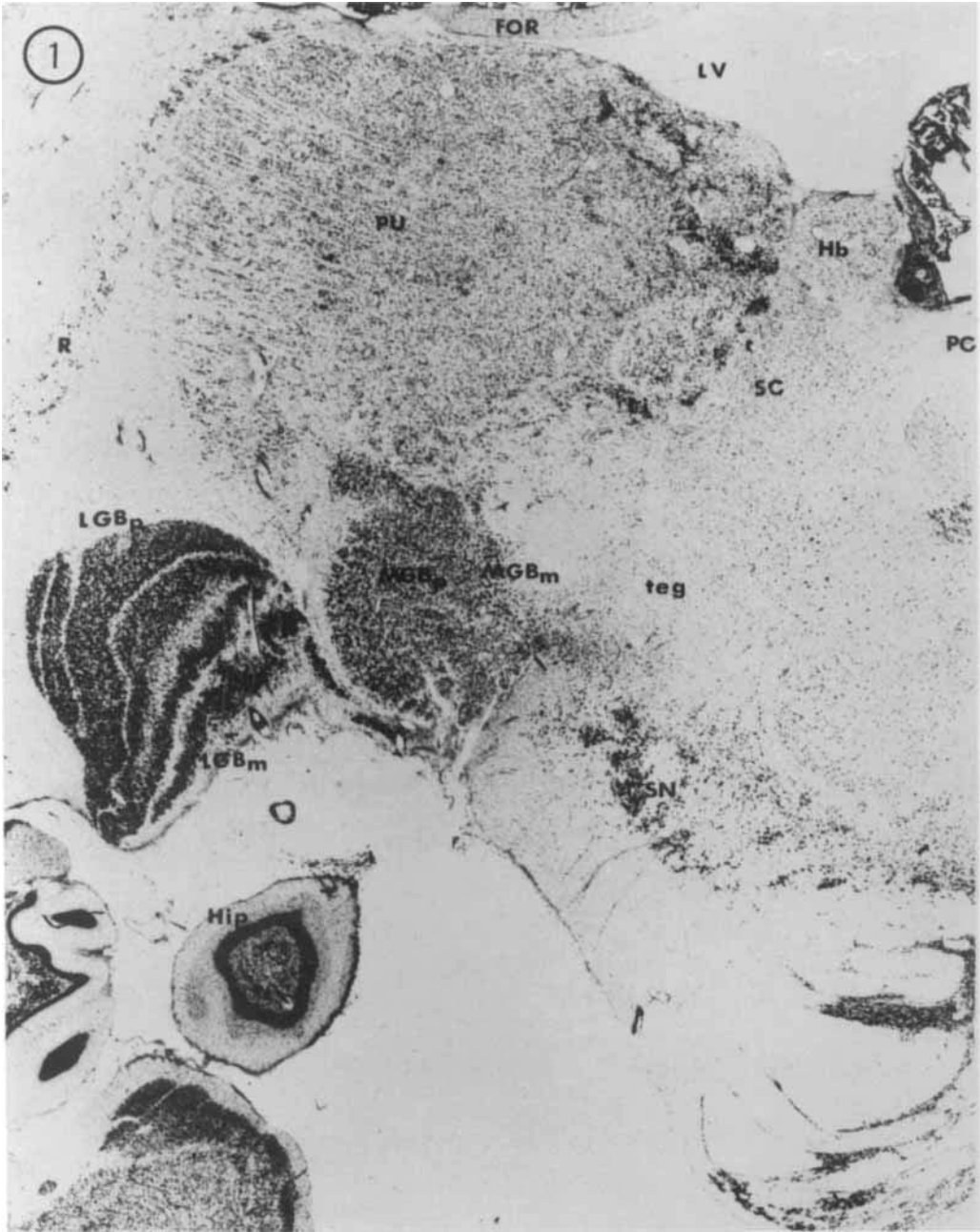


PLATE 2

EXPLANATION OF FIGURES

- 2 Horizontal section through gibbon lateral geniculate body.
- 3 Coronal section through chimpanzee thalamus. Arrows indicate probable location of a "barreloid." Arrowheads indicate possible "barreloid" (Van Der Loos, '76).

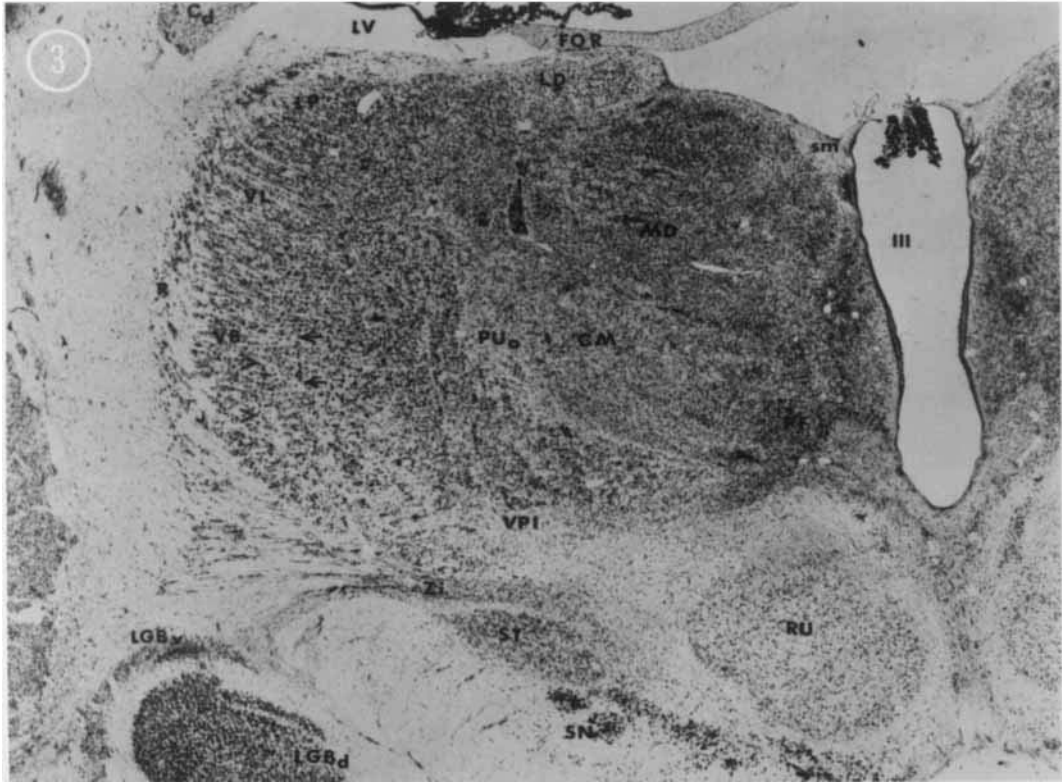


PLATE 3

EXPLANATION OF FIGURES

- 4 Relative frequencies of volumes of neuronal perikarya in the ventrobasal complex.
- 5 Relative frequencies of volumes of neuronal perikarya in medial geniculate body, pars parvocellularis (MBGp).
- 6 Relative sizes of specific sensory relay nuclei. Slopes (α) less than 1 indicate that the rate of increase in volume of these nuclei when comparing gibbons, great apes, and humans is less than that of the rest of the thalamus. The closer the line approximates a slope of 1, the more similar is the increase of the nucleus to that of the thalamus. The sensory relay nuclei have not only lagged behind the volumetric changes of the rest of the thalamus, but also differ among themselves. The lag is particularly prominent in the human specimens. Without these specimens, the slopes more closely approximate 1 ($VB\alpha = 0.99$, $r = 0.99$, $LGB\alpha = 0.71$, $r = 0.99$, $MGB\alpha = 0.92$, $r = 0.99$). Line fitted by least squares method.

